

```
!pip install FlagEmbedding
```

```
Requirement already satisfied: tqdm=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.19.0->FlagEmbedding)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets>=2.19.0->FlagEmbedding) (3.8.1)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.19.0->FlagEmbedding)
Requirement already satisfied: fsspec<=2025.3.0, >=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.6.0->FlagEmbedding)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.6.0->FlagEmbedding) (75)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.6.0->FlagEmbedding) (1.12.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.6.0->FlagEmbedding) (3.1)
Requirement already satisfied: jinj2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.6.0->FlagEmbedding) (3.1.6)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: tokenizers<=0.23.0, >=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: lxml>=4.5.2 in /usr/local/lib/python3.12/dist-packages (from ir-datasets->FlagEmbedding) (6.1.1)
Collecting lz4>=3.1.10 (from ir-datasets->FlagEmbedding)
  Downloading lz4-4.4.5-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (3.8 kB)
Requirement already satisfied: scikit-learn>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from sentence_transformers->FlagEmbedding)
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from sentence_transformers->FlagEmbedding)
Requirement already satisfied: aiohttp!=4.0.0a0, !=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: click<9.0.0, >=8.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.21.0->FlagEmbedding)
Requirement already satisfied: hf-xet<2.0.0, >=1.5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.21.0->FlagEmbedding)
Requirement already satisfied: httpx<1, >=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.21.0->FlagEmbedding)
Requirement already satisfied: charset_normalizer<4, >=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets>=2.19.0->FlagEmbedding)
Requirement already satisfied: idna<4, >=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets>=2.19.0->FlagEmbedding)
Requirement already satisfied: urllib3<3, >=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets>=2.19.0->FlagEmbedding)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets>=2.19.0->FlagEmbedding)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0->sentence_transformers->FlagEmbedding)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0->sentence_transformers->FlagEmbedding)
Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.6.0->FlagEmbedding)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinj2->torch>=1.6.0->FlagEmbedding)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.19.0->FlagEmbedding)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.19.0->FlagEmbedding)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.19.0->FlagEmbedding)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: multidict<7.0, >=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: yarl<2.0, >=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0, !=4.0.0a1->fsspec[http]<=2025.3.0->FlagEmbedding)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1, >=0.23.0->huggingface_hub>=0.21.0->FlagEmbedding)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1, >=0.23.0->huggingface_hub>=0.21.0->FlagEmbedding)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1, >=0.23.0->huggingface_hub>=0.21.0->FlagEmbedding)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=2.19.0->FlagEmbedding)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: pygments<3.0.0, >=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers<6.0.0, >=4.44.2->FlagEmbedding)
Requirement already satisfied: mdurl==0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers<6.0.0, >=4.44.2->FlagEmbedding)
Downloading flagembedding-1.4.0-py3-none-any.whl (247 kB)
  247.7/247.7 kB 23.1 MB/s eta 0:00:00
Downloading ir_datasets-0.6.3-py3-none-any.whl (947 kB)
  947.8/947.8 kB 48.5 MB/s eta 0:00:00
Downloading lz4-4.4.5-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (1.4 MB)
  1.4/1.4 MB 35.7 MB/s eta 0:00:00
Installing collected packages: lz4, ir-datasets, FlagEmbedding
Successfully installed FlagEmbedding-1.4.0 ir-datasets-0.6.3 lz4-4.4.5
```

```
from FlagEmbedding import BGEM3FlagModel

model = BGEM3FlagModel('BAAI/bge-m3',
                        use_fp16=True) # Setting use_fp16 to True speeds up computation with a slight performance degradation
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
 WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en

```

config.json: 100%                               687/687 [00:00<00:00, 60.1kB/s]

tokenizer_config.json: 100%                       444/444 [00:00<00:00, 41.6kB/s]

sentencepiece.bpe.model: reconstructing file: 100%          5.07MB / 5.07MB,   ???B/s

sentencepiece.bpe.model: downloading bytes:              3.62MB,   ???B/s

tokenizer.json: reconstructing file: 100%              17.1MB / 17.1MB, 1.70MB/s

tokenizer.json: downloading bytes:                    5.88MB,   ???B/s

special_tokens_map.json: 100%                      964/964 [00:00<00:00, 80.6kB/s]

Download complete: :      3.05GB, 66.5MB/s

Reconstruction complete: 100%    4.56GB / 4.56GB, 77.1MB/s

Fetching 30 files: 100%                               30/30 [00:50<00:00, 3.99s/it]

Loading weights: 100%                               391/391 [00:04<00:00, 1.12it/s]
```

#RECOMMENDATION SYSTEM MOVIES : BASED GENRE

```

movies = [
    {
        "title": "Inception",
        "genre": "Sci-Fi, Thriller",
        "description": "A skilled thief enters people's dreams to steal secrets and attempts to plant an idea in someone's mind
    },
    {
        "title": "Interstellar",
        "genre": "Sci-Fi, Adventure, Drama",
        "description": "Astronauts travel through a wormhole in space to find a new home for humanity."
    },
    {
        "title": "The Dark Knight",
        "genre": "Action, Crime, Drama",
        "description": "Batman faces a dangerous criminal mastermind who creates chaos and terror across Gotham City."
    },
    {
        "title": "Avengers: Endgame",
        "genre": "Action, Adventure, Sci-Fi",
        "description": "Superheroes reunite to reverse the devastating consequences caused by a powerful villain."
    },
    {
        "title": "Titanic",
        "genre": "Romance, Drama",
        "description": "A young couple from different social backgrounds fall in love during the tragic voyage of the Titanic."
    },
    {
        "title": "The Notebook",
        "genre": "Romance, Drama",
        "description": "Two young people fall deeply in love but face family pressure and separation before finding their way b
    },
    {
        "title": "The Conjuring",
        "genre": "Horror, Mystery, Thriller",
        "description": "Paranormal investigators help a family experiencing terrifying supernatural events in their new home."
    },
    {
        "title": "It",
        "genre": "Horror, Thriller",
        "description": "A group of children confronts a terrifying supernatural creature that appears in the form of a clown."
    },
    {
        "title": "The Hangover",
        "genre": "Comedy",
        "description": "Friends travel to Las Vegas for a bachelor party and wake up with no memory of what happened the previc
    },
    {
        "title": "3 Idiots",
        "genre": "Comedy, Drama",
        "description": "Three engineering students experience friendship, academic pressure, love, and the struggle to follow t
    },
]
```

```
{
  "title": "Dangal",
  "genre": "Sports, Drama",
  "description": "A former wrestler trains his daughters to become successful wrestlers despite social and family challenges."
},
{
  "title": "The Pursuit of Happyness",
  "genre": "Drama, Biography",
  "description": "A struggling father fights poverty while trying to build a better future for himself and his young son."
},
{
  "title": "The Matrix",
  "genre": "Sci-Fi, Action",
  "description": "A computer programmer discovers that reality is a simulated world controlled by intelligent machines."
},
{
  "title": "Jurassic Park",
  "genre": "Sci-Fi, Adventure, Thriller",
  "description": "Scientists create a theme park filled with genetically engineered dinosaurs, but the creatures escape and terrorize the visitors."
},
{
  "title": "Forrest Gump",
  "genre": "Drama, Romance",
  "description": "A kind-hearted man experiences important historical events while navigating friendship, love, and personal growth."
}
]
```

#embedding movies

```
texts = [
    f"{movie['title']}. Genre: {movie['genre']}. {movie['description']}"
    for movie in movies
]
```

```
for text in texts:
    print(text)
```

```
embedding=[]
for text in texts:

    embeddings= model.encode(text,
                              batch_size=12,
                              max_length=8192, # If you don't need such a long length, you can set a smaller value to speed up the process
                              )['dense_vecs']
    embedding.append(embeddings)
```

Inception. Genre: Sci-Fi, Thriller. A skilled thief enters people's dreams to steal secrets and attempts to plant an idea in someone's mind.

Interstellar. Genre: Sci-Fi, Adventure, Drama. Astronauts travel through a wormhole in space to find a new home for humanity.

The Dark Knight. Genre: Action, Crime, Drama. Batman faces a dangerous criminal mastermind who creates chaos and terror across Gotham City.

Avengers: Endgame. Genre: Action, Adventure, Sci-Fi. Superheroes reunite to reverse the devastating consequences caused by a previous event.

Titanic. Genre: Romance, Drama. A young couple from different social backgrounds fall in love during the tragic voyage of the Titanic.

The Notebook. Genre: Romance, Drama. Two young people fall deeply in love but face family pressure and separation before finding each other again.

The Conjuring. Genre: Horror, Mystery, Thriller. Paranormal investigators help a family experiencing terrifying supernatural events.

It. Genre: Horror, Thriller. A group of children confronts a terrifying supernatural creature that appears in the form of a clown.

The Hangover. Genre: Comedy. Friends travel to Las Vegas for a bachelor party and wake up with no memory of what happened the previous night.

3 Idiots. Genre: Comedy, Drama. Three engineering students experience friendship, academic pressure, love, and the struggle to find their own path.

Dangal. Genre: Sports, Drama. A former wrestler trains his daughters to become successful wrestlers despite social and family challenges.

The Pursuit of Happyness. Genre: Drama, Biography. A struggling father fights poverty while trying to build a better future for himself and his young son.

The Matrix. Genre: Sci-Fi, Action. A computer programmer discovers that reality is a simulated world controlled by intelligent machines.

Jurassic Park. Genre: Sci-Fi, Adventure, Thriller. Scientists create a theme park filled with genetically engineered dinosaurs, but the creatures escape and terrorize the visitors.

Forrest Gump. Genre: Drama, Romance. A kind-hearted man experiences important historical events while navigating friendship, love, and personal growth.

embedding

```
[array([-0.00202693,  0.0201381 , -0.01109634, ..., -0.05026658,
        -0.02285776, -0.01060469], dtype=float32),
 array([-0.02848639, -0.01454587, -0.01094803, ..., -0.02769613,
        -0.01036233, -0.01382132], dtype=float32),
 array([ 0.02084275, -0.00561649, -0.02002954, ..., -0.00626637,
        0.00443491, -0.00091101], dtype=float32),
```

```

array([ 1.7036576e-02,  6.5596098e-05, -2.2263847e-02, ...,
        9.5007792e-03, -1.9547462e-02, -4.3077352e-03], dtype=float32),
array([-0.01280117,  0.00654286, -0.02925876, ..., -0.00090848,
        -0.00543806, -0.03208977], dtype=float32),
array([ 0.0002628 ,  0.02121896, -0.02594645, ..., -0.02374038,
        -0.01331762, -0.02248945], dtype=float32),
array([-0.0235125 ,  0.00596215, -0.05109383, ..., -0.03574776,
        -0.01066542,  0.00150537], dtype=float32),
array([ 0.00691109,  0.03665004, -0.01400833, ..., -0.02990024,
        -0.0122845 , -0.00668951], dtype=float32),
array([ 0.01174693,  0.01989347, -0.01785168, ...,  0.02598397,
        0.00262097,  0.00159166], dtype=float32),
array([-0.01874067,  0.00562666,  0.00983806, ..., -0.00271972,
        -0.02530917, -0.02068929], dtype=float32),
array([-0.01804586, -0.00070537, -0.0304952 , ..., -0.0283718 ,
        0.00564185, -0.01977405], dtype=float32),
array([-0.02174005, -0.01807855, -0.02759006, ..., -0.03101528,
        0.01074827, -0.01955673], dtype=float32),
array([ 0.00279652,  0.01361282, -0.03184221, ..., -0.01509924,
        -0.00120035,  0.00162626], dtype=float32),
array([-0.022155 ,  0.00387926, -0.02752257, ..., -0.02850369,
        0.0029723 ,  0.0042977 ], dtype=float32),
array([-0.00415256,  0.0020796 ,  0.01115432, ..., -0.00900922,
        -0.03046259, -0.00698692], dtype=float32)]

```

```
#vector database:
```

```
!pip install -q chromadb
```

```
import chromadb
```

```
client= chromadb.Client()
```

```
collection= client.create_collection(name='movies_database')
```

```
len(embedding)
```

```
15
```

```
embedding[0]
```

```
array([-0.00202693,  0.0201381 , -0.01109634, ..., -0.05026658,
        -0.02285776, -0.01060469], dtype=float32)
```

```
ids = [f"movie_{i}" for i in range(len(embedding))]
ids
```

```

['movie_0',
 'movie_1',
 'movie_2',
 'movie_3',
 'movie_4',
 'movie_5',
 'movie_6',
 'movie_7',
 'movie_8',
 'movie_9',
 'movie_10',
 'movie_11',
 'movie_12',
 'movie_13',
 'movie_14']

```

```
collection.add(ids=ids,embeddings=embedding)
```

```
type(collection)
```

```
chromadb.api.models.Collection.Collection
def __init__(client: ClientI, model: CollectionModel, embedding_function:
Optional[EmbeddingFunction[Embeddable]]=DefaultEmbeddingFunction(), data_loader:
Optional[DataLoader[Loadable]]=None)
```

Abstract base class for generic types.

On Python 3.12 and newer, generic classes implicitly inherit from Generic when they declare a parameter list after the class's name::

```
class Mapping[KT, VT]:
```

```
collection.query(query_embeddings=embedding[0],n_results=1)
```

```
{'ids': [['movie_0']],
 'embeddings': None,
 'documents': [[None]],
 'uris': None,
 'included': ['metadatas', 'documents', 'distances'],
 'data': None,
 'metadatas': [[None]],
 'distances': [[0.0]]}
```